

E-commerce & Finance Insights Report

Generated via Python ReportLab • Author: Huzeif Khan

Key Performance Indicators (KPIs)

- **Total Revenue:** 10,641,558.95
- **Total Customers:** 4,338
- **Average Order Value:** 2,453.10

Visual Summary

Monthly Revenue Trend



Top 10 Products by Revenue

Top 10 Products by Revenue



Customer Segmentation (RFM Model)



Cohort Retention

Each cell shows the % of customers who returned after N months, grouped by their first purchase month (cohort).



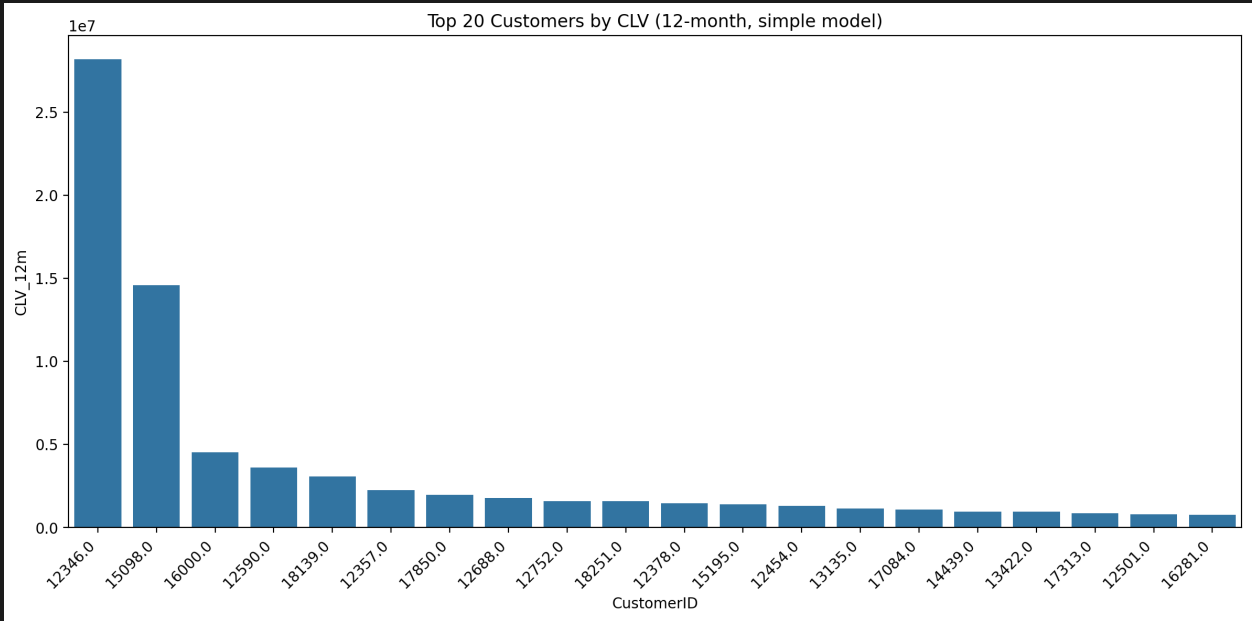
Revenue Forecast (Next 12 Months)

SARIMAX forecast with 80% confidence band. Use this to plan inventory, marketing budgets, and hiring.



Customer Lifetime Value (CLV) - 6-Month Outlook

We estimate a simple 6-month CLV as $AOV \times (\text{avg monthly repeat probability}) \times 6$, added to the historical revenue. Top 20 customers shown.



Customer Lifetime Value (CLV v1 - 12-Month Model)

Deterministic CLV v1: $AOV \times \text{Purchase Frequency per Month} \times 12 \text{ Months}$. Shows your top-value customers based on average order value and repeat purchase rate.



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CLV by Customer Segment

Average predicted 12-month CLV for each RFM segment.



RFM x CLV - Segment Insights

How lifetime value varies by customer segment, and how it correlates with Recency, Frequency and Monetary (RFM).

- **Regular Buyers** - avg CLV $\hat{a}$, -125,288 (n=1177)
- **At Risk / Lost** - avg CLV $\hat{a}$, -81,282 (n=882)
- **Loyal Customers** - avg CLV $\hat{a}$, -43,711 (n=1026)
- **Champions** - avg CLV $\hat{a}$, -14,341 (n=1253)

RFM-CLV Correlation



CLV vs RFM (example scatter)

